Linux

Definition of linux

Linux is an **open-source, Unix-like operating system** that manages hardware and software resources and allows users to interact through a **Command Line Interface (CLI)** or GUI.

Developed by **Linus Torvalds (1991)**

Key Features of Linux

- Open-source (free to use)
- Multi-user (many users at once)
- Multitasking (run multiple programs)
- Secure & stable
- Portable (runs on many devices)

Linux Architecture (Simple)

1. **Kernel** → Core of OS (controls hardware)
2. **Shell** → Interface between user & kernel
3. **File System** → Stores data
4. **Applications** → Software programs

Basic Linux Commands

File & Directory Commands

Command	Description
`pwd`	Show current directory
`ls`	List files
`ls -l`	Detailed list
`cd dir`	Change directory

Command	Description
`cd ..`	Go back
`mkdir dir`	Create folder
`rmdir dir`	Remove empty folder
`rm file`	Delete file
`rm -rf dir`	Delete folder

File Handling Commands

Command	Description
`touch file`	Create file
`cat file`	Display content
`less file`	View page by page
`head file`	First 10 lines
`tail file`	Last 10 lines
`cp file1 file2`	Copy
`mv file1 file2`	Move/Rename
`nano file`	Edit file

User Commands

Command	Description
`whoami`	Current user
`who`	Logged users
`id`	User ID info
`adduser name`	Create user
`passwd`	Set password
`su`	Switch user

Permissions Commands

Command	Description
`chmod`	Change permissions
`chown`	Change owner

Example:

```
chmod 755 file.sh
```

Process Management

Command	Description
`ps`	Show processes
`top`	Live processes
`htop`	Advanced view
`kill pid`	Stop process

Network Commands

Command	Description
`ping`	Check connectivity
`ifconfig`	Show IP
`ip a`	IP details (modern)
`netstat`	Network info
`ssh`	Remote login

Package Management

Amazon Linux / RHEL / CentOS

```
yum install httpd
```

Ubuntu / Debian

```
apt install apache2
```

Disk & Memory Commands

Command	Description
`df -h`	Disk usage
`du -sh`	Folder size
`free -m`	Memory usage

Important Command – `grep`

Used to **search text inside files**

Syntax:

```
grep "word" file.txt
```

Examples:

```
grep "error" logfile.txt
grep -i "linux" file.txt   # ignore case
grep -n "devops" file.txt # line numbers
```

Shortcuts

- `Ctrl + C` → Stop command
- `Ctrl + Z` → Pause
- `Ctrl + L` → Clear screen
- `history` → Previous commands

Interview Points

- Linux is **CLI-based OS**
- Kernel = **heart of Linux**
- Shell = **user interface**
- `grep` = **text searching tool**
- Widely used in **DevOps & Cloud**